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2

Data Preprocessing

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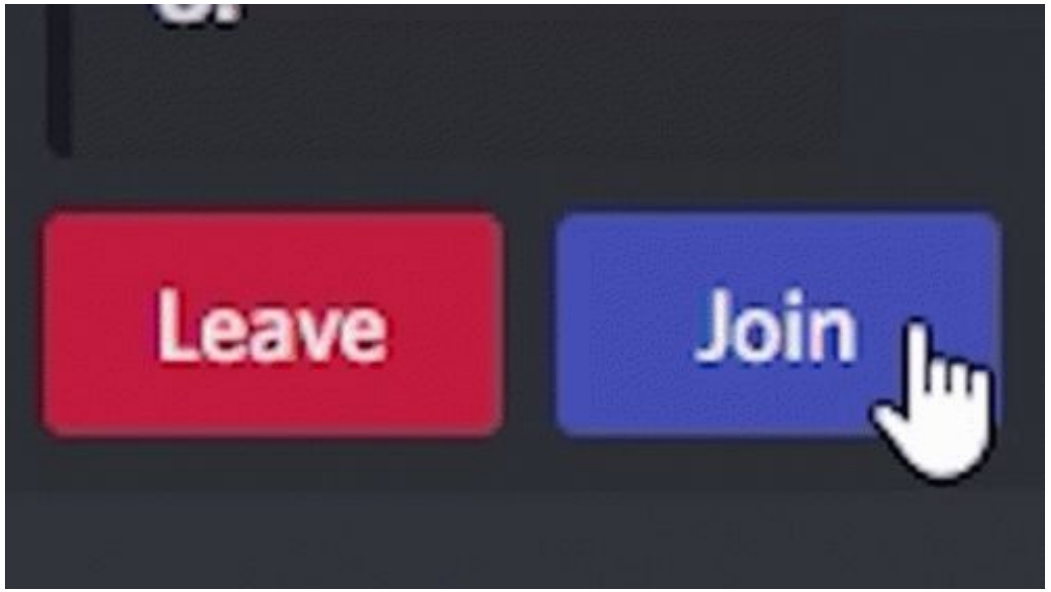

Quick Recap

Kindly add here...

Today's Agenda

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1. Data Ingestion

Loading data from CSV, Excel, APIs, and SQL databases effectively.

2. Data Inspection

Understanding structure, types, and statistical summaries.

3. Missing Data

Strategies for detection and imputation (Mean, Median, Mode).

4. Outliers

Detection using Z-Score & IQR, and handling techniques.

The Data Pipeline

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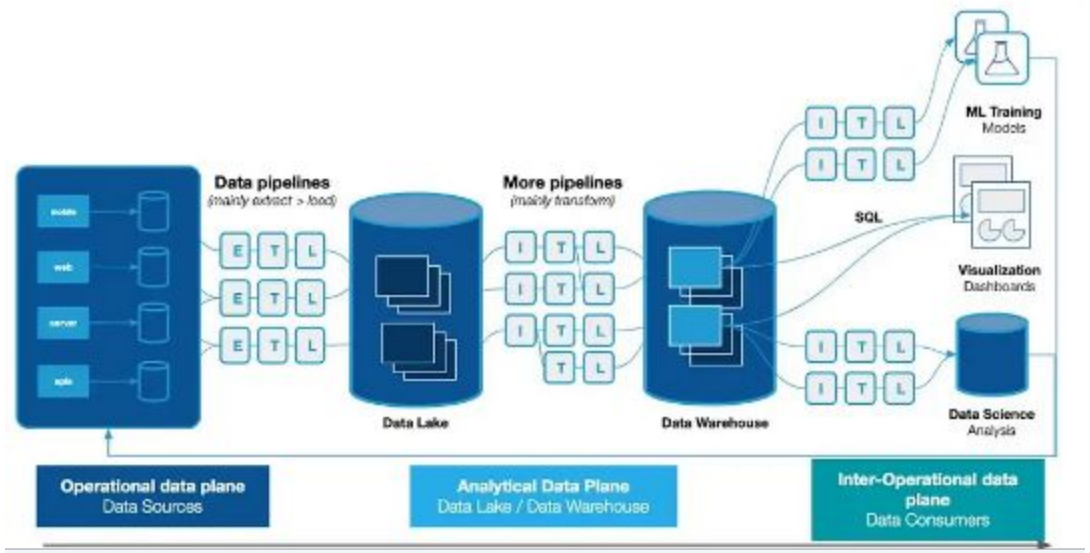
Garbage In, Garbage Out

Preprocessing is the most critical step in the ML pipeline. A model trained on dirty data will produce unreliable predictions.

Raw Data → Ingestion → Cleaning → Encoding → Model



Data Ingestion

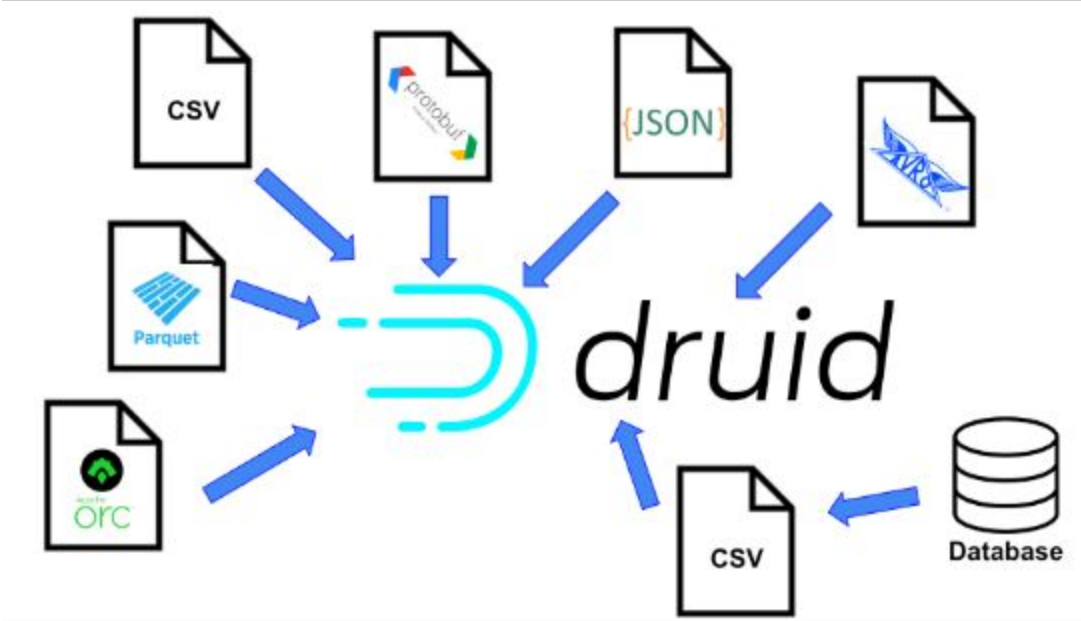
LOADING RAW DATA

What is Data Ingestion?

The process of transporting data from various sources to a storage medium or processing engine (like Pandas) for analysis.

Common Sources

- Flat Files: CSV, Excel, Parquet
- Databases: SQL, MongoDB
- Web: APIs, HTML Scraping



Reading Flat Files

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CSV

Comma Separated Values are the standard for data exchange.

```
import pandas as pd
df = pd.read_csv("data.csv")
```

Excel

Requires openpyxl library installed.

```
# Specify sheet name
df = pd.read_excel("data.xlsx",
    sheet_name="Sales")
```

Reading Web Data (APIs)

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JSON Format

APIs typically return data in JSON (JavaScript Object Notation), which Pandas handles natively.

We use the requests library to fetch the data first.

```python
import requests
import pandas as pd

# 1. Fetch data
url = "https://api.example.com/users"
response = requests.get(url)
data = response.json()

# 2. Convert to DataFrame
df = pd.DataFrame(data)
```

Reading from Databases

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SQL Integration

Pandas can execute SQL queries directly against a database connection.

Efficient for filtering data before loading it into memory.

```
import sqlite3
import pandas as pd

conn = sqlite3.connect("database.db")

query ="SELECT * FROM customers WHERE age > 25"

df = pd.read_sql(query, conn)

conn.close()
```

Ingestion Challenge: Encoding

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UnicodeDecodeError

Not all files are UTF-8. If you see "invalid start byte", it's usually an encoding issue.

```
# The Problem
df = pd.read_csv("file.csv")
# Error: utf-8 codec can't decode...

# The Solution
df = pd.read_csv("file.csv",
    encoding="latin1")
```

Ingestion Challenge: Large Data

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Processing in Chunks

If a file is larger than your RAM (e.g., 10GB CSV), you cannot load it all at once.

Use the chunksize parameter.

```
chunk_size = 10000
for chunk in pd.read_csv(
  "huge_file.csv", chunksize=chunk_size):

  # Process each chunk individually
  print(chunk.shape)
```

Data Inspection

EXPLORATORY DATA ANALYSIS

First Look at Data

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df.head(n)

Shows the first n rows. Essential to verify headers and data format.

df.tail(n)

Shows the last n rows. Good for checking footer data or truncation.

df.sample(n)

Returns n random rows. Great for checking data variety unbiasedly.

Structure & Types

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df.info()

The most powerful summary method. It provides:

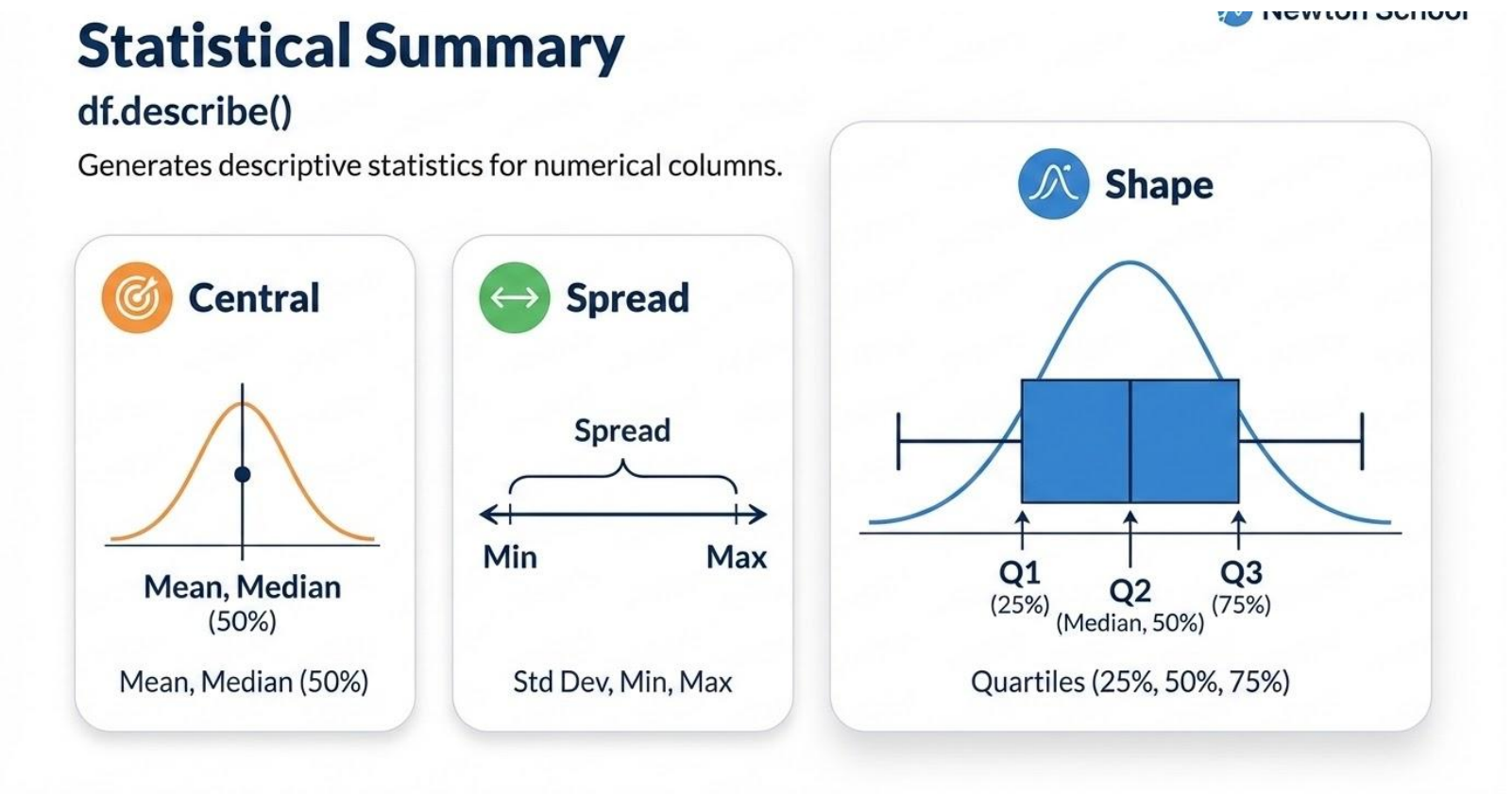
- Total Number of Rows & Columns
- Column Names & Data Types
- Non-Null Counts
- Memory Usage

```
RangeIndex: 1000 entries, 0 to 999
Data columns (total 3 columns):
# Column Non-Null Count Dtype
--- ------ -------------- -----
0 Age 950 non-null float64
1 City 1000 non-null object
2 Salary 980 non-null float64
```

Statistical Summary

df.describe()

Generates descriptive statistics for numerical columns.



Missing Data

IMPUTATION STRATEGIES

Why is Data Missing?

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MCAR

Missing Completely At Random

No pattern. e.g., A lab sample was dropped.

MAR

Missing At Random

Missingness depends on observed data. e.g., Test scores missing for absentees

MNAR

Missing Not At Random

Missingness depends on the value itself. e.g., High earners not disclosing income.

Visualizing Missingness

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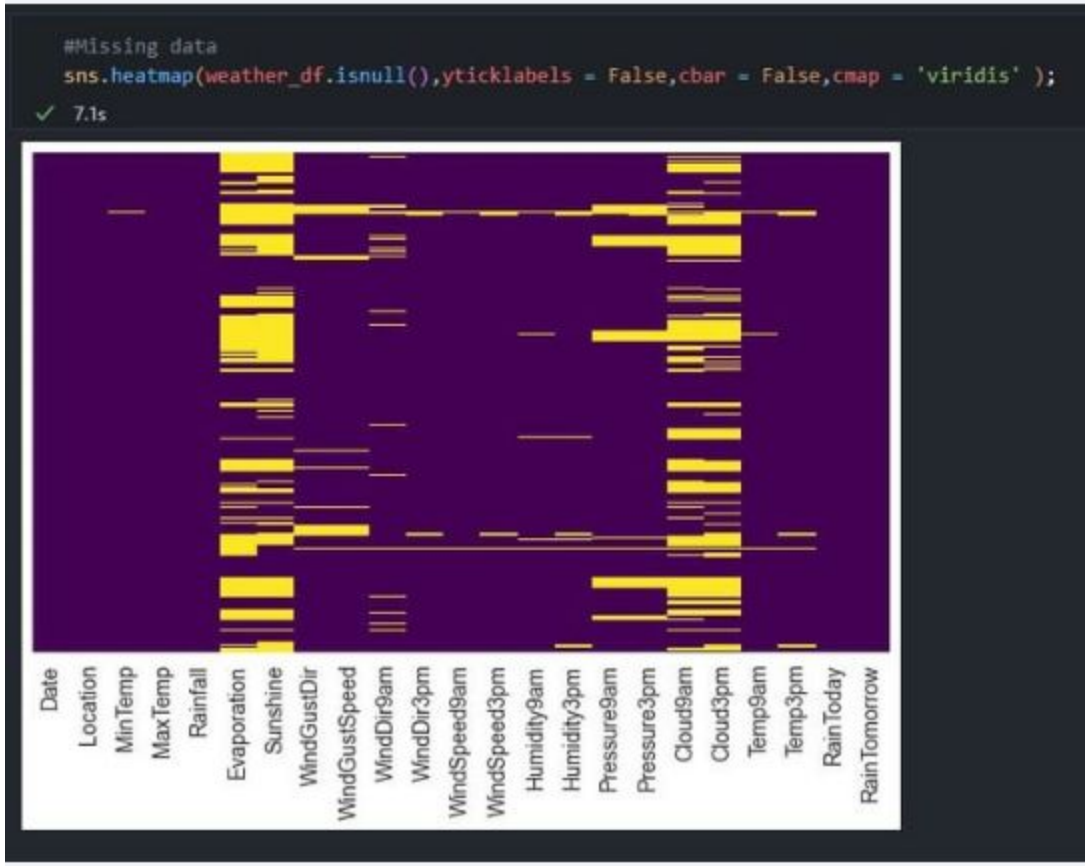
Seaborn Heatmap

A heatmap is the best way to see patterns in missing data. White/yellow lines often indicate missing values.

```
import seaborn as sns
sns.heatmap(df.isnull(), cbar=False)
```



Strategy 1: Deletion

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Row Deletion

Use when missing data is < 5% and random.

```
df.dropna(axis=0)
```

Column Deletion

Use when a column has > 60% missing data.

```
df.dropna(axis=1)
```

Strategy 2: Mean Imputation

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Concept

Replace NaN with the average of the column.

- Best for: Normally distributed numerical data.
- Risk: Sensitive to outliers.

```
# Fill missing Age with Mean
mean_age = df['Age'].mean()

df['Age'].fillna(mean_age,
  inplace=True)
```

Strategy 3: Median Imputation

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Concept

Replace NaN with the 50th percentile value.

- Best for: Skewed data (Income, House Prices).
- Benefit: Robust against outliers.

```
# Fill missing Salary with Median
med_sal = df['Salary'].median()

df['Salary'].fillna(med_sal,
  inplace=True)
```

Strategy 4: Mode Imputation

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Concept

Replace NaN with the most frequent value.

Best for: Categorical Data (City, Gender).

```
# Fill missing City with Mode
mode_city = df['City'].mode()[0]

df['City'].fillna(mode_city,
  inplace=True)
```

Strategy 5: Time Series Filling

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In time-series data (e.g., Stock Prices), values depend on previous days.

Forward Fill (ffill)

Propagate the last valid observation forward.

```
df.fillna(method='ffill')
```

Backward Fill (bfill)

Use the next valid observation to fill gap.

```
df.fillna(method='bfill')
```

Handling Outliers

DETECTION & REMOVAL

What are Outliers?

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Definition

Data points that deviate significantly from other observations. They can be errors (Age=200) or valid extreme cases (Billionaire in salary data).

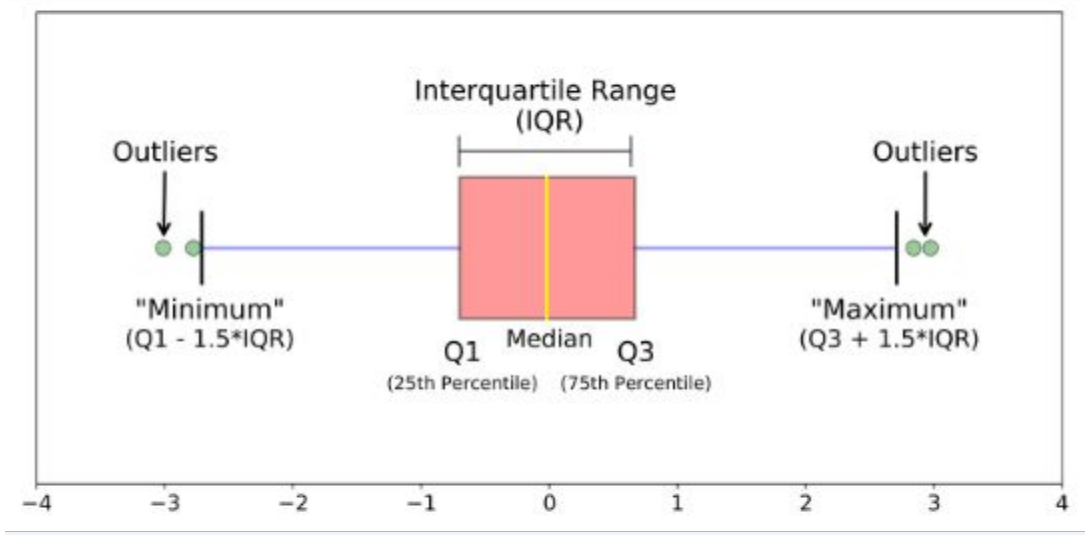
Impact

Outliers pull the Mean towards them, distorting analysis. Algorithms like Linear Regression are highly sensitive to them.

Visualizing: The Boxplot

The Boxplot is the standard tool for outlier detection.

- **Box:** Contains 50% of data (IQR).
- **Whiskers:** Extend to 1.5 * IQR.
- **Dots:** Points outside whiskers are outliers.



Method 1: Z-Score

Concept

The Z-score tells you how many standard deviations a data point is from the mean.

If Z > 3 or Z < -3, it is considered an outlier.

$$Z = \frac{X - \mu}{\sigma}$$

Z-Score Implementation

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```python
from scipy import stats
import numpy as np

# Calculate Z-scores for a column
z_scores = np.abs(stats.zscore(df['Salary']))

# Filter: Keep only rows where z < 3
df_clean = df[z_scores < 3]

print("Removed outliers:",
  len(df) - len(df_clean))
```

Method 2: IQR (Interquartile Range)

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Concept

More robust than Z-score because it uses quartiles, not mean/std dev.

Any data point outside the bounds is flagged as an outlier.

IQR = Q3 (75th%) - Q1 (25th%)

Lower Bound = Q1 - 1.5 * IQR

Upper Bound = Q3 + 1.5 * IQR

IQR Implementation

```
Q1 = df['Salary'].quantile(0.25)
Q3 = df['Salary'].quantile(0.75)
IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

# Filter
df_clean = df[(df['Salary'] ≥ lower_bound) &
              (df['Salary'] ≤ upper_bound)]
```

Handling Strategies

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Trimming

Completely removing the outlier rows.

Pro: Simple.

Con: Data loss.

Capping

Replacing outliers with the upper/lower limit (Winsorizing).

Pro: Preserves data size.

Con: Modifies distribution.

Encoding

CATEGORICAL TO NUMERICAL

Categorical Data Types

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Ordinal Data

Categories with an inherent order.

- Low, Medium, High
- Satisfied, Neutral, Dissatisfied
- Sizes: S, M, L, XL

Nominal Data

Categories with no order.

- Red, Green, Blue
- New York, Paris, Tokyo
- Dog, Cat, Bird

Method 1: Label Encoding

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Use for Ordinal Data

Assigns a unique integer to each category.

Caution: Do NOT use for Nominal data (e.g., Apple=1, Banana=2). The model will think Banana > Apple.

```
from sklearn.preprocessing import
LabelEncoder

le = LabelEncoder()
df['Size_Encoded'] =
le.fit_transform(df['Size'])

# S=0, M=1, L=2
```

Method 2: One-Hot Encoding

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Use for Nominal Data

Creates a binary column for each category.

drop_first=True prevents multicollinearity (Dummy Variable Trap).

```
df = pd.get_dummies(df,
  columns=['Color'],
  drop_first=True)
```

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Q&A

Thank You!